

INFO345: Research Topics in Recommender Systems

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Abstract

This project presents a hybrid recommendation system designed to incorporate users' emotional states in addition to their listening preferences. Traditional recommendation systems typically rely on content-based similarity or collaborative filtering, but they often overlook the emotional impact of music. By combining audio feature analysis, collaborative filtering of listening behaviour, and a mood-based scoring system, our model aims to recommend songs that align with both the user's musical taste and desired emotional experience. We preprocess datasets from the Million Song Dataset [1] and the Echo Nest Taste Profile Subset [20], engineer a mood classification system using the circumplex model of emotions [12], and implement a hybrid recommender system. Evaluation demonstrates the potential of mood-aware recommendations, while highlighting limitations such as the lack of real user mood feedback and ethical considerations regarding emotional influence.

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1 Introduction

Music is a universal source of happiness, comfort, and inspiration and is an important factor that can influence our mood and emotional state. From a stressful day to giving energy during a work out, music can often help us connect to our emotions. With today's most popular streaming platforms, like Spotify and Apple Music, we often get recommendations based on listening patterns and genre preference. While these methods are effective, they often lack the ability to capture one of the most fundamental dimensions and music. Its emotional resonance.

This project aims to develop a music recommendation system that puts the emotional state of the user at the centre. By shifting the focus from only "what do you like" to "how do you feel" and "how would you like to feel?", we aim to create a more personalised, relevant and inspiring experience. We will implement a hybrid recommendation system that combines three components. First, a content-based component that uses low-level audio features to compare similar songs. Second, a collaborative filtering component that captures patterns in users' actual listening behavior. Finally, we will add mood into the recommendation, allowing the system to adjust similarities based not only on audio features and listening history, but also on a person's current emotional state.

The main goal is to make a system that not only gives you suggestions on music you have liked, but also music that will affect you in the way you want to feel. This will add a new dimension to music recommendations and might increase the user engagement and the quality of the listening experience.

2 Data Analysis

When finding datasets for our model, we looked at multiple big datasets from Spotify. However, most of them did not include user information and would thus limit our possibilities for creating a nuanced and hybrid model. We ended up with two connected datasets: one including low-level song features (such as valence, tempo, energy, etc.), and one including user-item information (user_id, track_id, and playcount). This way, we could combine the data from both datasets to create a hybrid recommendation system which looks at the listening history of the user, as well as audio features.

2.1 Description of Data

The main dataset we chose is a reconstructed version of The Million Song Dataset [1]. This dataset combines data from lastfm-spotify-tags-sim-userdata [11], lastfm-dataset-2020 [17], tagtraum genre annotations [15], and the Spotify API. The user-item dataset is a dataset from The Echo Nest Taste Profile Subset [20] containing real user data. Both datasets contain a `track_id`, so we can combine the data from them.

The datasets contain detailed information on 50683 songs and a comprehensive user listening history with over 9.7 million records. In the subsection below, you can find a description of each column.

Since our original dataset did not include a "mood" feature, we decided to create our own. This allows us to implement a hybrid recommendation strategy where mood predictions are derived from audio features through content based filtering, and collaborative filtering personalizing final recommendations. This way, the system takes both mood of track and individual user preferences into account. The process for generating this mood feature is described more in detail in the preprocessing section below.

Both datasets can be downloaded from: <https://www.kaggle.com/datasets/undefinnull/million-song-dataset-spotify-lastfm>.

2.1.1 Column Description of Music Information Dataset. The track dataset provides basic identifiers to connect tracks to user interaction (`track_id`, `spotify_id`), along with *name*, *artist*, *genre*, and *year*. However, its most important role in our system lies in the Spotify *audio features*, which allow us to estimate the emotional qualities of music. Features such as *valence* (musical positivity), *energy* (perceived intensity), *danceability*, *tempo*, and *mode* are all linked to affective responses to music and therefore form the basis of our emotion mapping. For example, high *valence* and *energy* often correspond to cheerful tracks, while lower values are associated with sadness or melancholy. Additional features such as *acousticness*, *speechiness*, and *instrumentalness* provide nuance by differentiating between acoustic ballads, spoken-word/rap, or instrumental music[19].

- **track_id**: ID of track from Million Song Dataset [1].
- **name**: Name of song.
- **artist**: Main artist of song.
- **spotify_preview_url**: Direct links to 30 second audio preview of songs on Spotify in mp3 files[11].
- **spotify_id**: Spotify ID used to query Spotify API [11].
- **tags**: Genre tags from lastfm dataset [11].
- **genre**: Genre tags from tagtraum genre annotations [15].
- **year**: Release year of track.
- **mood**: Custom feature added to dataset which is not present in the original datasets mentioned above. It represents the overall emotional tone of song inferred

from audio features and is divided into four quadrants, which are described in detail later in the paper.

The following column descriptions for audio features are based on the Spotify API documentation.[19].

- **duration_ms**: The track length in milliseconds.
- **danceability**: Measurement of how easy it is to dance to a song depending on things like tempo, rhythm and how steady the beat is. Ranging from 0.0 (not danceable) to 1.0 (very danceable).
- **energy**: Measures energy of song depending on how fast and loud it is. For example, death metal has high energy while a Bach prelude is low energy. Measured from 0.0 to 1.0.
- **key**: Represents musical key of song. Number correspond to notes using the Pitch Class notation.
- **loudness**: Describes how loud song is in decibels (dB) which are then averaged across the entire song. Values typically fall into the range of -60 and 0 db.
- **mode**: Describes if a song is in a major or minor key. 1 represents major key and 0 is minor key.
- **speechiness**: Measures how much spoken content a song has. Values above 0.66 usually mean the song is mostly speech such as an audiobook. Most songs have values between 0.33 and 0.66, and means there is a mix between speech and music. Values below 0.33 have little to no speech.
- **acousticness**: Measure of how acoustic a song is ranging from 0.0, not confident in acousticness, to 1.0 highly confident of acousticness.
- **instrumentalness**: Estimates if a track has no vocals. Sounds like "ooh" and "aah" count as instrumental. Values above 0.5 are usually considered to be instrumental, with greater confidence as the value gets closer to 1.0.
- **liveness**: Detects if there is an audience present in the song. Values above 0.8 indicate strong likelihood that the track is live.
- **valence**: Describes a song's musical positivity on a scale between 0.0 and 1.0. Higher values sound happy or cheerful, while low values sound sad or negative.
- **tempo**: Describes an estimation of song's beats per minute (BPM).
- **time_signature**: Describes how many beats are in each measure ranging from 3 to 7. For example, a 3/4 time signature means 3 beats per measure.

2.1.2 Column Description of User Listening History Dataset. The second dataset captures user behaviour with `user_id`, `track_id`, and `playcount`. This enables a collaborative component, where the model considers which tracks are enjoyed by other users with similar listening patterns.

- **track_id**: ID of track from Million Song Dataset.

- **user_id**: Unique identifier of each user.
- **playcount**: Number of times a user has listened to the song.

2.2 Data Analysis & Preprocessing

We began by importing and inspecting both datasets to gain an overview of the structure and contents.

2.2.1 Preprocessing and Visualization of Music Info Dataset. For the Music Info dataset, the initial preprocessing involved dropping unnecessary or redundant columns to remove irrelevant information and reduce noise. The **name column** (song title) was dropped as each song already has a unique identifier through track ID, as well as that encoding it would create thousands of non-informative features, so therefore including it would only increase model complexity without adding predictive value. However, we store this column as metadata separately for displaying song recommendation to a user after model predicts relevant track IDs. The columns **spotify_preview_url** and **spotify_id** do not provide any meaningful information since we already have a unique identifier through **track_id**, and was therefore also dropped.

We then checked for missing values and duplicates to maintain data quality and consistency. A missing data visualization (Figure 1) was generated to illustrate the completeness of the dataset.

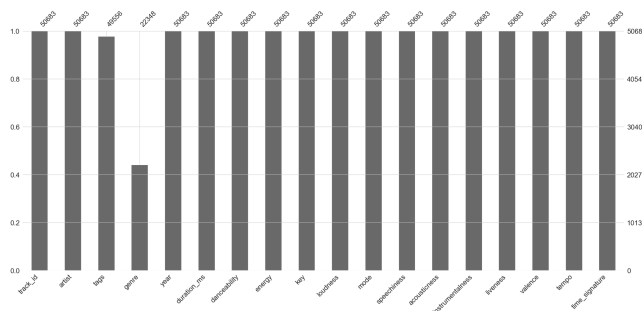


Figure 1: Missing Values in the Music Info Dataset

We decided to drop the **genre column** due to the large proportion of missing values. Additionally, as the **tags column** already contains information about genres, removing the genre column will simplify the dataset without losing critical information.

Next, categorical variables such as **artist** and **tag** were encoded into numerical representations. Each song has multiple tags, so we treat them as multi-hot features using scikit-learn’s MultiLabelBinarizer. Each unique tag becomes a separate column that we mark 1 if the song has the mentioned tag, and otherwise 0. This allows the model to learn which tag combinations are associated with user preferences. For the

artist column, we use label encoding because each track in our dataset has a single artist, so there is no need to expand into multiple columns like one-hot encoding. Label encoding will assign a unique integer to each artist.

Numerical audio features such as energy, tempo and loudness were normalized to a standard scale, preventing features with large numerical ranges from dominating the analysis. A correlation heatmap (Figure 2) was produced to visualize relationships between these features, revealing patterns such as a strong positive correlation between energy and loudness, and weaker or inverse correlations among emotional and acoustic attributes.

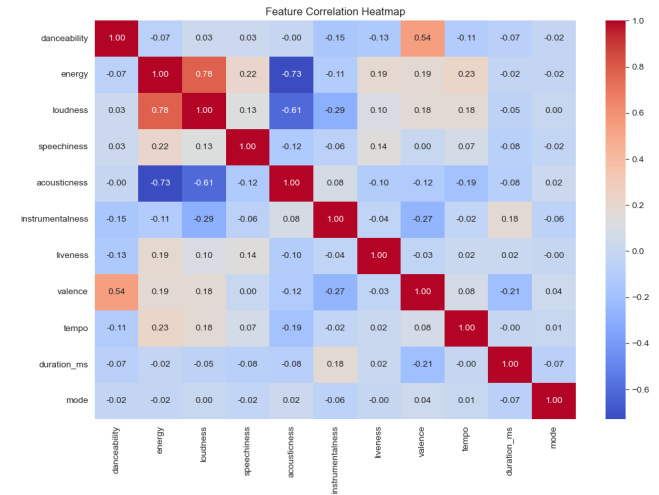


Figure 2: Correlation Heatmap for Audio Features

After normalization, we used the pandas DataFrame describe() to inspect feature averages (Table 1), which made it easier to categorise them into emotions. For example, suppose a happy song is recognisable by high valence, energy, danceability and positive mode. In that case, we can use this information to categorise it in a way where songs with a positive mode and valence, energy and danceability above the median or 75 percentile is categorised as happy.

In addition, mood classification was performed to categorize tracks into four emotional quadrants (Q1-Q4) based on their audio features. Later in the preprocessing these quadrants were encoded using the scikit-learn’s MultiLabelBinarizer, which converts the quadrants into four separate binary columns, where a value of 1 indicates the song belongs to that quadrant, otherwise 0. The resulting mood distribution plots (Figure 3) highlighted the relative prevalence of different emotional tones within the dataset. A more detailed description of the quadrants and the criteria used for classification is provided in the music and mood section below.

Table 1: Descriptive Statistics for Song Features

	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo	mode
count	50683.000000	50683.000000	50683.000000	50683.000000	50683.000000	50683.000000	50683.000000	50683.000000	50683.000000	50683.000000
mean	0.500544	0.686486	0.812495	0.079688	0.214666	0.225509	0.215640	0.436187	0.516996	0.631060
std	0.181377	0.251808	0.071468	0.079672	0.304065	0.337387	0.184881	0.260604	0.123992	0.482522
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.369168	0.514000	0.779752	0.036897	0.001406	0.000018	0.098498	0.215509	0.421453	0.000000
50%	0.504057	0.744000	0.829641	0.050524	0.040060	0.005636	0.138138	0.407855	0.510639	1.000000
75%	0.629817	0.905000	0.862811	0.087526	0.341365	0.441441	0.289289	0.638469	0.592892	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

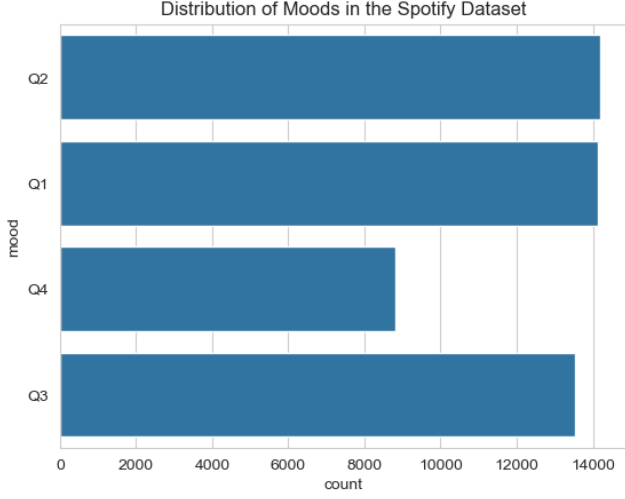


Figure 3: Mood Distribution

We also examined tag correlations and computed the average emotional feature per tag (Figure 4), looking at how the top 15 tags correspond to particular moods.

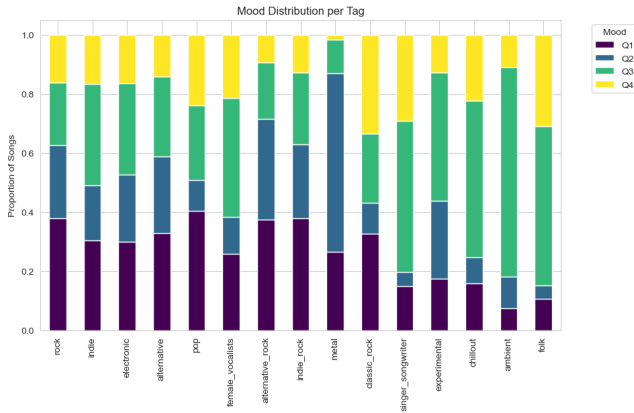


Figure 4: Mood Distribution Per Tag

2.2.2 Preprocessing and Visualization of User Listening History Dataset. For the User Listening History dataset, we explored the distribution of playcounts (Figure 5) to understand overall listening behaviour among users.

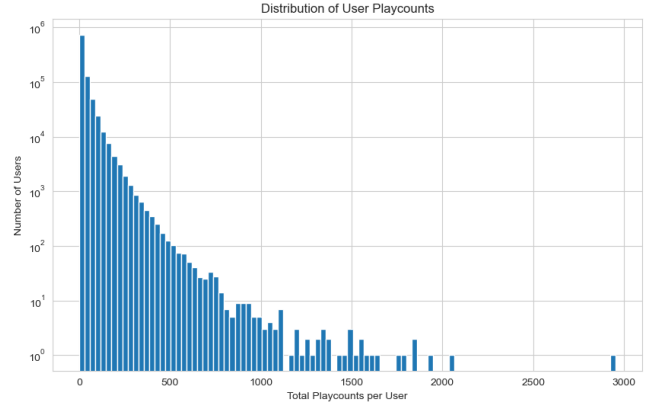


Figure 5: Distribution of User Playcounts

The distribution plot above shows a long-tail pattern, where a small subset of tracks has a high number of plays, while most songs had relatively few listens. We decided to drop users who have listened to less than 50 songs as they will not be providing enough valuable data for our recommendation model. To handle outliers, we decided to put an upper limit of 2000 playcounts per user. Despite most playcounts between 1500-2000 belonging to only a single user, we treat them as natural outliers and keep them in the dataset, as there is a big chance it reflects high engagement and genuine listening behavior.

Similarly to the music info dataset, we also checked for missing values and duplicates, although there were none in this dataset. Furthermore, to ensure consistency across the datasets, we verified that the track_id column matched correctly between the User Listening History and Music Info datasets.

Overall, preprocessing and visualizations made up a good foundation for model development. The cleaned data, along with knowledge gained from visualizations, allows for more

reliable and interpretable outcomes in the subsequent stages of the recommendation model and analysis process.

3 Music and Mood

Music has an extraordinary ability to influence how we feel. One single song can energize us, calm us or evoke different emotions. Music affects us subconsciously, and often in a positive way. Have you ever watched a movie without any music or any sound? Have you ever thought about what emotions a song gave you in a specific scene or moment? Have you ever been to a mall or store without music in it? Or have you ever been to any event without music? In our daily life we feel, experience and sense the music when doing everyday tasks. Even though the feeling of music is individual for every person, every person who lives in today's fast fast-paced world listen to music at some time during the day.

Music has never been as accessible as it is today; there are a bunch of free music platforms. And even the ones you pay for are not so costly. According to Kramer [10], users have never had so much control over the music that is playing as they have now. We can now, to a much larger degree, decide what we want to play, when we want to play, what songs to skip and it what context we want to listen to music.

Music and mood have an extraordinary connection to each other, and an interesting study done on driving and music [21] found that music with different energy level gives significant mood changes and even in your driving. The close connection between emotion and music is not accidental; it rather reflects a complex interaction between musical structures, human perception, culture and personal experience. [2] This research highlights the central question:

What in the music triggers specific emotions and responses? And do these vary across different individuals, cultures and time?

We will discuss some of these aspects further in the Ethical Issues and Concerns section.

3.1 Mood Generator

When recommending songs based on mood we needed to categorize each song into different moods. These classifications are rooted in research in psychological and data science about how we perceive music. The relationship between music and mood is a complex and dynamic field. And as mentioned in the article [2] Music Emotion Classification: A Literature Review

Music often evokes a wide range of emotions, and a single piece of music can express a combination of feelings.

You might feel a different range of emotions; furthermore, emotions are not static but rather dynamic. Research done by Frank et al. [5] mention that the interpretation of music can be affected by culture or time. This means the artist and composers might intend one emotion (emotion X), but today listener may experience something else than the intended emotion (emotion Y). Moreover our emotional response is individual and may be affected by biology, culture and memories. Even though there are a lot of different challenges, there exist some research that show us that there are some universal musical features that affects emotions for modern Western listeners. This made the different classifications easier.

Table 2: Emotions and features

Features	Typical emotional effect
Major vs. minor mode	Major → happy; Minor → sad or serious
Tempo	Fast → energetic; Slow → calm or sad
Rhythm	Strong/stable → power, joy; Smooth/fluid → calmness
Pitch	High pitch → bright, joyful; Low pitch → heavy, solemn
Timbre	Bright timbre → excitement; Warm/dark timbre → melancholy
Harmony / dissonance	Consonance → safety, happiness; Dissonance → tension, fear

In practice, assigning moods to different songs require a combination of theoretical knowledge and practical heuristics. To assign each track to an emotional category, we created a rule-based mood classification system inspired by research [5] on musical emotion renegotiation. The model uses normalized audio features such as valance, energy, loudness, acousticness, mode and danceability. Each features was scaled so the major differences wouldn't make too much of an impact. The classification rules were developed based on well-established emotional dimensions. And was mainly based on Table 4 in the article Music and Emotion[5]. Because the features do not align perfectly with one single emotion category the classification must be understood as a heuristic approximation rather than an absolute truth. Music and emotions are subjective, and over time, these features can change. As mentioned in Music Emotion Classification: A Literature Review [2], most emotion datasets are biased towards Western music. So even different cultures and events can give a person a whole new perception of what the song is. The model also allows for multiple moods to be assigned if several rules apply, reflecting that music often conveys blended or overlapping emotional qualities rather than one single emotion. If no rule is satisfied the track will be labeled neutral, which then prevents forced or incorrect classifications.

When building the mood generator we discovered some limitations while using the model from [7]. While using this model the majority of our songs where not sorted into any catagory, due to our specific sorting. Therefore we decided

to redo the model and base it on the mapping on emotions from Frank et al.[5]. The quadrants used for our sorting is pictured below in Figure 8.

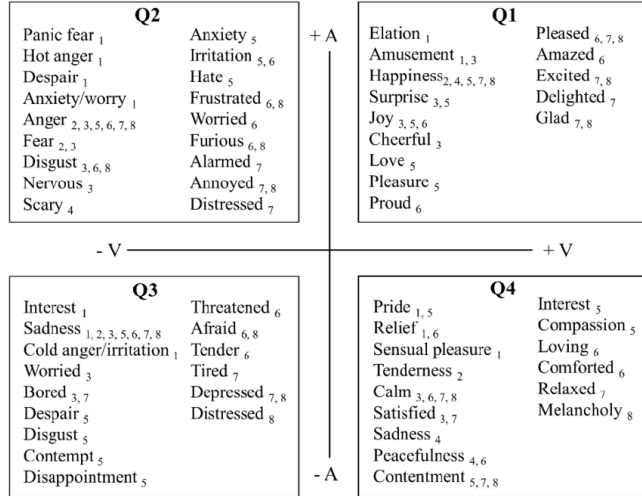


Figure 6: Mapping of selected emotion words [5]

While this model is much simpler it did however sort the majority of the music into different categories as seen above in Figure 3. Furthermore this model is also based on music psychology and the circumplex model of emotion. This model is often used when organizing emotions into different groups. The circumplex model of emotion is a two dimensional model that sorts emotions based on arousal and valence. Valence measures how positive or negative an emotion is (x-axis). High valence = positive and happiness and low valence = negative and sadness. Arousal on the other hand measures how energetic or intense an emotion is. High arousal = high energy, low arousal = low energy, calmness and so on. By combining valence and arousal, we get four main quadrants, each representing a group of related emotions as shown below.

Q1 High Valence and High Arousal (Top Right) This is a positive and energetic quadrant.

- Feelings that are both pleasant and intense
- *Examples of feelings/music:* joy, excitement, euphoria, happiness
- *Music example (typical):* Fast tempo, major mode, high volume

Q2: Low Valence and High Arousal (Top Left) This is the negative and energetic quadrant

- Feelings that are unpleasant, but intense and highly activated
- *Examples of feelings/music:* Anger, fear, rage, stress, anxiety

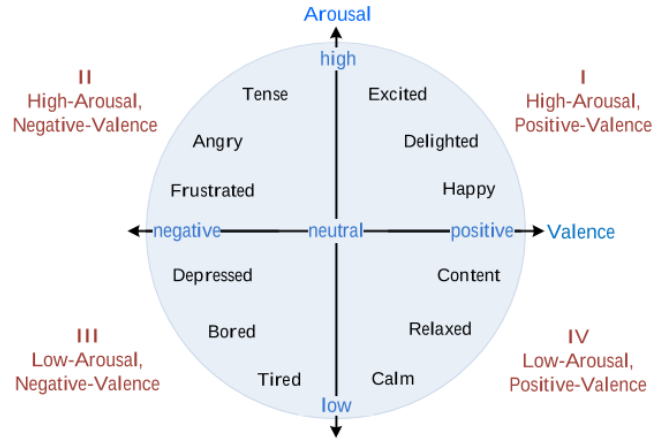


Figure 7: Circumplex-Model [12]

- *Music example (typical):* Fast tempo, minor mode, dissonance, high volume

Q3: Low Valence and Low Arousal (Bottom Left) This is the negative and calm quadrant

- Feelings that are unpleasant and deactivated (low in energy)
- *Examples of feelings/music:* Sadness, depression, boredom, exhaustion, melancholy
- *Music example (typical):* Slow tempo, minor mode, low volume

Q4: High Valence and Low Arousal (Lower Right) This is the positive and calm quadrant

- Feelings that are pleasant, but relaxed and low in energy
- *Examples of feelings/music:* Calm, contentment, relaxation, peace, ceremonial
- *Music example (typical):* Slow tempo, major mode, soft sounds, steady rhythm

This model sorted the majority of our songs. As there may be some limitations due to the simplicity of the sorting the conclusion was that this was the most sufficient for us for this project.

4 Description of Recommendation Model

When choosing music, there are so many options that you can't even wrap your head around them. Do you filter them by artist, genre, tempo or maybe you just listen to the top 100 songs on your music provider? To solve this problem, recommendation systems come to play. Here users will get a unique experience, with unique features and music based on their own taste and style. Recommender systems aim to give you a personal recommendation and rating based on what the model thinks you would rate the item. This may help the user find new items that the user otherwise would not have

found on their own. The system is trained to understand the preferences based on interactions.

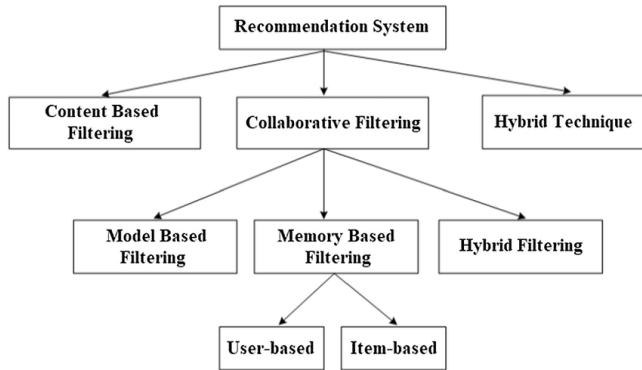


Figure 8: Enter Caption

4.1 Why Hybrid Model?

In our recommender system, we combine content-based and collaborative filtering techniques in a hybrid model combined with calculations of mood score, as it leverages both low-level song features (which we used to infer mood) and the user's actual listening behaviour. This gives us a way to recommend music that matches a user's current emotional state, or the emotional state they want to reach, even for songs the user have never heard before.

The collaborative filtering component ground the recommendations in real user behaviour: we know which songs they have listened to, how often, and how other users with similar listening patterns behave. This provides personalization beyond pure emotional similarity. For instance, two songs may be emotionally similar, but one may be more likely to be well-received because it is also popular among users with similar listening history. By combining the content-based and collaborative filtering components, we both mitigate the cold-start problem, where new or rare songs get emotionally scored via content, and capture patterns in what people really choose to listen to, via collaborative signals.

It is important to acknowledge known limitations from the start. Listening history does not necessarily indicate user preference - a song might appear frequently in a user's history simply because it is part of a shared playlist or because friends or family members use the same account. For example, a younger child might listen to children's songs while older siblings or parents listen to more mature tracks. In such cases, the data does not accurately reflect the preference of any one individual. This limitation highlights the importance of combining content-based information (which directly models the musicality of the songs) with collaborative signals (which provide broader content about user

engagement) to create recommendations that are more robust, personalized, and emotionally relevant.

Importantly, because our system is emotion-aware, the hybrid model gives us additional nuance: the content-based part measures how similar songs are based on audio features, while the collaborative part captures user preferences, and a mood scoring method that evaluates whether a song's labeled mood matches a user's current mood. This way, recommendations are not just emotionally "correct" in theory, but emotionally relevant and likely to be engaging for that specific user, given their history and broader listening community. In short, the hybrid approach helps us balance emotional matching, personalization, diversity, and robustness.

4.2 Building the Hybrid Song Recommender

When building our recommender model, we began by importing both the cleaned datasets from the preprocessing section as well as a file containing metadata of songs, such as artist name and song name to make displaying recommendations easier than giving a track ID.

To enable content-based recommendations, we selected the relevant numerical audio features from the dataset such as danceability, energy and tempo. In the original Kaggle notebook by Shivarov [18] which we took inspiration from, the author uses an Annoy index for the content-based recommender to find similar tracks. Annoy is an approximate nearest neighbors library which "splits the vector space into multiple random projection trees to build an index" [4], and is used for very fast similarity searches on large datasets. In our implementation, we chose to use k-Nearest Neighbors (kNN) from scikit-learn instead. kNN performs exact similarity searches, and is often simpler to implement. The main trade-off is between speed and exactness, where Annoy is faster but approximate, whereas kNN is exact but can be slower on large datasets. Since we are already more familiar with kNN, we found it easier to use and sufficient for our dataset size. We then trained a k-Nearest Neighbors model using the numerical audio features of each track. The model uses cosine similarity to compare songs and identify those that are the most similar, in this case the 50 closest matches for any given song, forming the basis of our content-based recommender. Further, given the track ID, we made a function that queries the kNN model and returns the indices of the top n most similar tracks, excluding the track itself, along with the corresponding distances which we will use later as similarity scores for **content-based filtering (CB) recommendations**.

Due to collaborative filtering relying on a user-item matrix, both users and tracks must be represented with integer indices so the model can find matrix positions. Therefore

we added the forward and reverse mapping dictionary code written by Shivarov [18] that translate between original ID and assigned integer indicies. This enables fast lookup in both directions and keeps data consistent across both collaborative and content-based parts of the hybrid recommender in progress.

Next, we convert the user-track playcount data, functioning as our user-item matrix, into a sparse matrix, following the method used by Shivarov [18]. Here we also apply a Truncated SVD, which according to Khadka, decomposes the matrix into its singular values and vectors and then selects top k singular values and vectors [8]. In our code, this decomposition results in user factors and item factors, where the dot product of these matrices estimates the original playcounts. These are then used to generate **collaborative filtering (CF)** recommendations.

The **mood scoring component** of the hybrid recommendation system adds another level of personalization. Each song in the dataset was previously labeled with a mood quadrant, and the goal is that the user can specify their mood when requesting recommendations. For each song, we check whether its mood matches the user’s mood and assigns a score of 1 for a match and a 0 for a mismatch.

To combine these approaches, we normalize the CF-, CB-, and mood scores to be comparable. The final recommendation score for each song in the hybrid recommender is a weighted combination of these scores, inspired by Seddik’s lecture on Hybrid-Knowledge-Based Recommender systems [16], where weighting parameters called beta values allow us to adjust the influence of the different recommendation techniques. The outcome is a ranked list of recommended tracks that balances a user’s listening preferences, similarities to a chosen song, and a user’s current emotional state.

5 Evaluation and Discussion

The addition of mood as a weighted component in the hybrid recommendation model allows the system to better personalize recommendations. By combining collaborative filtering, content-based similarity and mood scoring, our hybrid recommender can suggest songs that reflect user’s listening history and their current emotional state. While mood scoring in some ways are comparable to content-based similarity, since the mood labels in our dataset are inferred from audio features, it emphasizes the overall emotional quality of a song rather than specific low-level audio features like tempo or instrumentality. Instead of directly comparing low-level audio features like tempo or instrumentality, the mood scoring groups songs according to overall mood. Our thought behind this was that the system will recommend tracks that matches user’s current emotional state, even if songs differ in specific audio feature details.

5.1 Evaluation Metrics

The evaluation metrics reported in Table 3 are standard ranking metrics used for recommender systems [3]. *Precision@K* measures the proportion of top- k recommended items that are relevant for the user (in their listening history), which captures how accurate the model is at suggesting them correctly. *Recall@K* measures how many of a user’s relevant items were successfully retrieved in the top- k , meaning it evaluates how complete the recommendations are. *Hit Rate@K* is a binary measure indicating whether at least one relevant item appears in the top- k results. *NDCG@K* (Normalised Discounted Cumulative Gain) additionally considers the ranking of the items by giving higher scores to relevant tracks appearing at the top of the list, thus measuring not only correctness but also how well a system orders its recommendations. Finally, *MAP@K* (Mean Average Precision) represents the average quality of rankings across all relevant recommendations and across all users, rewarding models that consistently place relevant tracks in high-ranked positions.

When inspecting the evaluation metrics for the recommendation models, the collaborative filtering model stands out as the best-performing approach overall. It scores higher on all ranking-based metrics, indicating that it is more successful at recommending items that users have previously consumed. However, it is important to interpret these metrics with caution. The evaluation here measures well the models can predict that a user has *listened to a song before*, rather than whether they enjoyed it or would want to listen to it again. This distinction is important, as user listening behaviour can be heavily driven by common playlists and trending artists. Consequently, collaborative filtering tends to perform well simply because users with similar behaviour often listen to the same popular tracks.

The hybrid model incorporates mood-based features but uses randomly assigned moods in the current evaluation. Since there is no real user feedback regarding mood preferences, the model cannot be meaningfully evaluated. To properly assess the hybrid approach, the system would require feedback indicating whether mood-aware recommendations align with their preferences. Without such data, traditional metrics cannot capture the value of personalization introduced by mood-aware recommendations.

5.2 Baseline Models

To run a further analysis of our model we made some baseline models and compared them to our own model (Table 4). To make our baseline model we decided to make one based on completely random metrics, a random recommender baseline. This is a good reference to how our model does compared to pure coincidence. If our model had not been able to beat this

Table 3: Evaluation metrics for recommendation models at $k = 5$

Model	Precision@5	Recall@5	Hit Rate@5	NDCG@5	MAP@5
Hybrid: CF:0.2, CB:0.2, Mood:0.6	0.0258	0.0088	0.1125	0.0754	0.0050
CF	0.0568	0.0195	0.2264	0.1409	0.0100
CB	0.0003	0.0001	0.0015	0.0009	0.0001
Hybrid: CF:0.4, CB:0.2, Mood:0.4	0.0321	0.0111	0.1344	0.0899	0.0063
Hybrid: CF:0.7, CB:0.3, Mood:0.0	0.0360	0.0123	0.1510	0.1045	0.0074

score we could have concluded that our model did not work. Compared to the user's list of top songs based on playcount, the model predicted a quite low percentage.

When we analyzed our model to the popularity baseline our result were as expected a lot better then the random recommender model. Popularity recommenders are often known to be quite good and often have far better results. This will in some way be a good comparison to our own model and can be a realistic comparison for us. When comparing them we used hit rate to show the relevance and the recommender quality in the models. As you can see in the result table below they were not to unexpected. A major limitation is the songs that are not played at all. They will not be sorted. Moreover, you can see that the model for random had no hits similar to any other recommender.

On further notice we decided to not use MSE and RSME, this is due to its limitation and what makes sense to use for this project. If we had used RMSE and MSE this would make sense for the CF model, but the metric this provides would not give us results for the hybrid model and CB model.

5.3 Ethical Issues and Concerns

Music is in many ways a source of happiness and even has had a documented impact on your mood. Music can offer many different benefits, like reduced stress, joy, relaxation, and empowerment [22]. Perhaps the most important benefit is that music participation is often associated with emotional regulation and can often be a strategy to help regulate and manage mood. [22] But can it be harmful as well? As mood based recommendation systems grow more and more sophisticated, they might raise concerns about emotional manipulation. [14] Music psychology is a term used for courses that study how people experience, feel, learn and use music. They also investigate how it affects us and is, therefore, an important field in our recommender system and something we have to take into consideration.

According to Manning Children's Center [13] music can have a very harmful effect on people, especially if the person already dwells on on negative feelings. It can even lead to depression, anxiety or make an already existing condition worse. When teens listen to sad music when they already feel sad they might become even more sad. When the music

is used to feel understood in tough times it might increase the symptoms. If our system is used by these kinds of people is it really justifiable to release for people. Do we exclude some and then include some. Do we never recommended sad songs? While music can be used to support emotional regulation it can also reinforce "bad" emotions, because our system makes the used choose its mood first this might be an ethical issue we have to take into consideration.

Another issue that we have to take into consideration is the mood-classifications. We use simplified rules [6], and they may not reflect how listeners actually perceive mood. The classification may be wrong, and if these mood labels are used to influence users emotional state it can lead to an unintentional mood manipulation. A user trying to relax may instead be pushed towards music that increase stress and may lead to negative side effects. When making this model it is important to consider the real life scenarios and impacts it might have on people. Another important impact that we have to take into consideration is addictiveness. What if the user misused the system. They want to use the music to not feel certain emotions and never process certain emotions. The majority of the studies already mentioned show that people use music to regulate stress, sadness, anger, energy and other emotions. *Music as an Emotion Regulation Strategy* (MERS) [21] [9] is a method used and Koelsch who wrote Brain correlates of music-evoked emotions show us what part of the brain that is activated and what they trigger. This sets the stage for music to act as self-medication, because the reward system responds. With all these concerns, is it justifiable for us to make the system and does the good outweigh the bad?

5.4 Limitations & Possible Improvements

Despite the potential of our hybrid music recommender, several limitations should be noted. First, a high playcount does not necessarily indicate that a user likes a track. Some songs may appear frequently due to shared playlists or multiple users on the same account, rather than reflecting individual preference. Including explicit user ratings or feedback could improve recommendation accuracy.

Table 4: Hit score for random user

Track (Artist)	CF	CB	Hybrid	Random	Popular	Hit Count
Help I'm Alive (Metric)	1		1		1	3
I'll Be Your Man (The Black Keys)	1		1			2
Quando o Sol se For (Detonautas Roque Clube)		1	1			2
Go On, Say It (Blind Pilot)		1	1			2
Everlasting Light (The Black Keys)	1					1
Gimme Sympathy (Metric)	1					1
Remember When (Side A) (The Black Keys)	1					1
Can You Stand The Rain (New Edition)		1				1
Maggie May (Rod Stewart)		1				1
Bye Bye Baby (Bay City Rollers)		1				1
Ayo Technology (Milow)			1			1
Rival (Pearl Jam)				1		1
Monday, Monday (The Mamas & the Papas)				1		1
Survival Of The Sickest (Saliva)				1		1
The Northern Cobbler (Crippled Black Phoenix)				1		1
Every Home Is Wired (Porcupine Tree)				1		1
MIA (Chevelle)					1	1
Revelry (Kings of Leon)					1	1
Such Great Heights (The Postal Service)					1	1
Float On (Modest Mouse)					1	1

Second, the current mood classification is simplified into four quadrants based on valence (positivity) and arousal (energy). While this captures general emotional characteristics, it cannot fully represent complex or overlapping emotions in music. Future work could explore more nuanced mood labelling or apply machine learning models to predict mood directly from audio features.

Third, evaluation of the mood-aware recommendations is limited because the mood labels were generated heuristically, without real user feedback. Collecting responses from users regarding emotional relevance would allow for a more meaningful assessment of the hybrid system.

Fourth, emotional responses to music vary across cultures and individuals. Our model relies on generalised patterns that may not reflect all user experiences. Incorporating cultural context or self-reported emotional preferences could enhance personalisation.

Finally, mood-aware recommendations raise ethical considerations. Users attempting to regulate their emotions could be exposed to music that reinforces negative feelings or fosters emotional dependency. Implementing safeguards, such as user controls or content warnings, could mitigate potential negative effects.

Addressing these limitations would improve both the effectiveness and the ethical deployment of mood-aware music recommendation systems.

6 Summary of Findings

Our project on creating a hybrid emotion-aware music recommendation system shows that emotional context has great potential to add value to music recommendations. Our hybrid recommendation system uses listening patterns through collaborative filtering, song similarity through content-based filtering and mood input to give more relevant recommendations. For our model, collaborative filtering performs strongest across precision, recall, hit rate and ranking quality. Content-based filtering performs weaker on its own, but seems to strengthen the hybrid scores when combined with collaborative filtering. The mood component adds an extra dimension of personal preference, yet standard ranking metrics do not capture this effect because it only tests whether the user listened to the track before. The hybrid recommendations show stable results when weights shift toward collaborative filtering, while mood heavy weights drop in performance due to lack of real user mood data.

The model still faces limits in mood labels, missing explicit user ratings and variation in how listeners across cultures interpret emotion in music. Ethical concerns stay important because mood based recommendations influence how users feel. Our hybrid recommender forms a decent foundation for future work where refined mood labels, explicit user feedback, improved evaluation data, and user protection features improve accuracy and safety.

We give permission for our report to be used as an example in the future.

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